

BUS TICKETING SYSTEM USING RFID,SERVO AND LCD

1st GANGADHAR RAO G.P.

*Dept.of ECE, Qis College Of Engineering And Technology
Vengamukkapalem (523272) ongole,Andhra Pradesh
poornagangadhar3@gmail.com*

3rd AJAYA REDDY VADICHARALA.

*Dept.of ECE, Qis College Of Engineering And Technology
Vengamukkapalem (523272) ongole,Andhra Pradesh
ajayarreddyvadicharala@gmail.com*

5th YEDULLA HUSSEN SHAIK.

*Dept.of ECE, Qis College Of Engineering And Technology
Vengamukkapalem (523272) ongole,Andhra Pradesh
shaikyedullahussen053@gmail.com*

2nd PUJITHA MUTHYALA.

*Dept.of ECE, Qis College Of Engineering And Technology
Vengamukkapalem (523272) ongole,Andhra Pradesh
muthyalapujitha@gmail.com*

4th NAVYA BHARGAVI KANCHARLA.

*Dept.of ECE, Qis College Of Engineering And Technology
Vengamukkapalem (523272) ongole,Andhra Pradesh
navyabhargavi15@gmail.com*

6th VENKATA AJAY KUMAR MIRIYALA.

*Dept.of ECE, Qis College Of Engineering And Technology
Vengamukkapalem (523272) ongole,Andhra Pradesh
mvenkataajay123@gmail.com*

Abstract—Today, it is very difficult to buy tickets in city buses because of the large number of people trying to buy tickets from the conductor in queues. So in order to overcome the situation, we propose an automated fixed bus fare(debeted) system.using the Rfid card and Arduinio uno.The cards being resuable,they are much more convenient compared to the paper ticketing system. An RFID reader scans the passenger's card when tapped, and the Arduino Uno microcontroller processes the data from the card. If the card is valid and contains sufficient balance, a fixed fare amount is deducted, and the updated balance is displayed on a 16x2 LCD. The ISD1820 voice module provides an audible confirmation of a successful fare deduction, while a servo motor acts as an automated gate, allowing entry only after payment. In case of invalid cards or insufficient balance, a buzzer generates an alert and the gate remains closed.

Index Terms—Arduinio uno,Rfid reader,Rfid card/tags,Servo motor,Buzzer,ISD1820 module,Lcd

I. INTRODUCTION

Public transportation plays a crucial role in the daily commute of millions of passengers. However, traditional bus ticketing systems still rely heavily on manual fare collection, which is prone to human errors, delays, and revenue leakage. Manual systems also require additional manpower and result in slower passenger boarding times, especially during peak hours. With the increasing demand for smart city infrastructure, there is a strong need to automate fare collection to make public transport faster, more efficient, and more reliable. Automated Fare Collection (AFC) systems using contactless technologies are already being adopted in metros and premium bus services in many cities. Among various technologies, Radio Frequency

Identification (RFID) stands out as a cost-effective and reliable solution for identifying passengers quickly and securely. RFID eliminates the need for cash transactions or paper tickets and allows seamless passenger entry with a simple tap on the card.

This paper presents the design and implementation of an RFID-based fixed fare bus ticketing system. The system uses an RFID reader to detect passenger cards and an Arduino Uno microcontroller to process data and deduct a fixed fare amount. A 16x2 LCD display provides visual feedback, while an ISD1820 voice module gives audio confirmation. A servo motor acts as an automated entry gate, and a buzzer alerts in the event of invalid or insufficient balance cards. The proposed system is low-cost, scalable, and easy to implement. By automating fare collection, it reduces boarding time, prevents revenue loss, and improves passenger convenience. The system can be deployed in buses, trams, and small-scale public transport systems, making it a step toward smart transportation solutions for future cities.

II. IMPLEMENTATION APPROACH

To implement this project in a systematic way, the following steps were carried out:

III. COMPONENT SELECTION AND DATA COLLECTION

The first stage involved identifying the components that are suitable for the project. The main component of this project is the RFID reader (RC522), which operates at 13.56 MHz and is used to detect RFID tags. It was selected because it is low-cost, has a simple interface with Arduino via SPI communication,

Identify applicable funding agency here. If none, delete this.

and provides reliable short-range card reading. RFID tags or cards are passive devices that store passenger identification data and can be reprogrammed with balance information. The Arduino Uno is chosen as the main microcontroller due to its simplicity, open-source nature, and availability of libraries for RFID, LCD, and servo motor interfacing. Processes card data, deducts the fare, and triggers other output devices. The 16x2 LCD display is selected for its ability to clearly display passenger details, fare deductions, and error messages. For access control, a servo motor is used to act as a gate controller. A buzzer is added to provide audible alerts for successful and failed transactions, while the ISD1820 voice module is chosen to give a human-friendly voice confirmation of each card tap.

Data collection involves gathering information on typical fare structures, average passenger boarding time, and common problems in existing ticketing systems. This data is used to set appropriate fare amounts, card balance limits, and system response times.

IV. SYSTEM DESIGN AND LOGIC FLOW

The design consists of four main functional blocks: input, processing, output, and power supply. The input block includes the RFID reader (RC522) that detects the unique ID of the RFID card when tapped by the passenger. This data is sent to Arduino Uno via SPI communication for processing.

The processing block is handled by the Arduino Uno microcontroller, which stores the fixed fare amount in its program memory. Upon receiving card data, the Arduino compares it with pre-stored card IDs and checks for validity. If the card is authorized, the fare amount is deducted, and the updated balance is stored.

The output block provides visual, auditory, and mechanical feedback to the user. The 16x2 LCD displays messages such as 'Fare Deducted' or 'Insufficient Balance.' The ISD1820 voice module plays a pre-recorded audio message confirming the transaction. The servo motor acts as a gate controller, opening only if the fare is successfully deducted, while the buzzer gives warning tones for invalid transactions. The power supply block ensures that stable 5V DC is provided to all modules for smooth operation.

The logic flow of the proposed RFID-based fixed fare bus ticketing system begins with powering the Arduino-based circuit and initializing all components, including the RFID reader, LCD display, servo motor, buzzer, and voice module. Once the system is ready, it continuously scans for an RFID card near the reader. When a card is detected, the Arduino reads its unique ID and verifies whether it is registered. If the card is valid, the system checks for sufficient balance. In case of insufficient balance or an invalid card, the buzzer is activated, a warning message is displayed on the LCD, and the gate remains closed. If the card is valid and the balance is sufficient, the system deducts a fixed fare amount from the stored balance, updates the balance in memory, and displays the updated value on the LCD. The voice module plays a confirmation message to notify the passenger. During

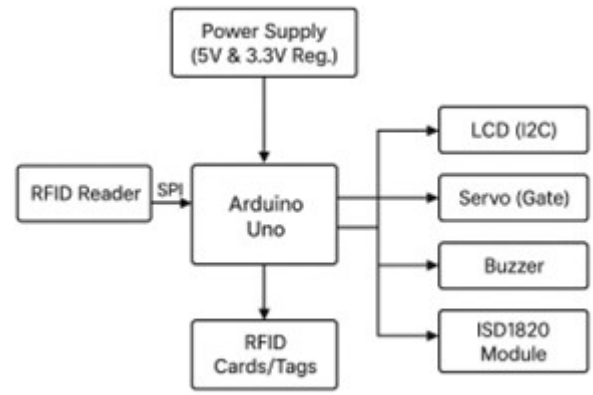


Fig. 1. BLOCK DIAGRAM

the same time, the servo motor rotates to open the entry gate for a few seconds, allowing the passenger to enter, and then automatically returns to the closed position. After this, the system resets and waits for the next card detection, thus maintaining a continuous automated ticketing cycle.

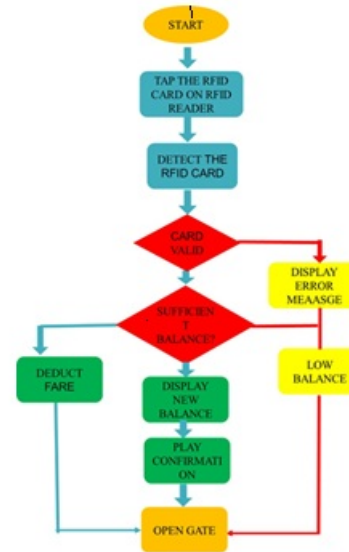


Fig. 2. LOGIC FLOW

V. SOFTWARE DEVELOPMENT PHASE

The software development process begins with the selection of the programming environment. For my project, the Arduino IDE is used because it provides a simple interface for writing, compiling, and uploading code to the Arduino Uno. Open-source libraries such as (MFRC522) for RFID communication, (Servo.h) for motor control, and (LiquidCrystal.h) for LCD display are included to simplify implementation and reduce development time.

The program is structured into modular functions for better readability and maintenance. The `setup()` function initializes the RFID reader, LCD, servo motor, buzzer, and voice module. This ensures all components are ready before the main logic begins execution. The `loop()` function continuously waits for an RFID card to be detected. When a card is tapped, its unique ID is read and compared with pre-stored IDs in the program memory.

If a valid card is detected, the program checks if the card balance is greater than or equal to the fixed fare value stored in a global variable. If sufficient balance is available, the balance is updated by deducting the fare amount and a success message is displayed on the LCD. The servo motor is activated to open the gate for a specified duration, and the ISD1820 voice module plays a pre-recorded 'Ticket Accepted' message. The buzzer remains silent in this case.

If the card is invalid or the balance is insufficient, the program triggers the buzzer, displays an error message on the LCD, and keeps the gate closed. A short delay is used before returning to the main loop to wait for the next card.

VI. REAL-TIME MONITORING AND DISPLAY PHASE

The real-time monitoring and display phase is a crucial stage in the proposed RFID-based fixed bus fare system, as it enables passengers and operators to receive immediate feedback regarding fare transactions, system status, and passenger authentication. This phase ensures transparency, reduces confusion, and improves the overall efficiency of the ticketing process.

A. RFID TAGS

The RFID tag acts as the passenger's identity card. It contains a microchip with a unique identification number (UID) and an antenna for communication with the RFID reader. Passive RFID tags are used in this project as they do not require a battery and derive power from the reader's electromagnetic field. Each tag corresponds to a passenger's account, from which the fixed fare is deducted during bus boarding. Tags are durable, reusable, and provide secure, contactless operation.

When the card is tapped near the RFID reader, the reader captures the tag's UID and sends it to the Arduino Uno for processing. Based on the stored logic, the system deducts a fixed fare from the passenger's balance and provides confirmation via the LCD, buzzer, and ISD1820 voice module. Since each RFID tag has a globally unique identifier, there is minimal risk of duplication or fraud, ensuring secure transactions. The RFID tag offers several advantages in public transportation applications, including contactless operation, durability, and reusability. It speeds up the ticketing process by eliminating manual cash handling and reduces congestion during passenger boarding.



Fig. 3. RFID TAGS

B. RFID READER

The RFID reader acts as the communication bridge between the RFID tag and the Arduino Uno. It generates an electromagnetic field that activates passive tags and receives the transmitted UID. The reader then sends this unique code to the microcontroller for validation. In this project, the reader is directly connected to the Arduino through serial communication pins, ensuring fast and accurate data transfer. Its ability to quickly detect and authenticate tags plays a key role in ensuring smooth passenger entry and reducing boarding delays.

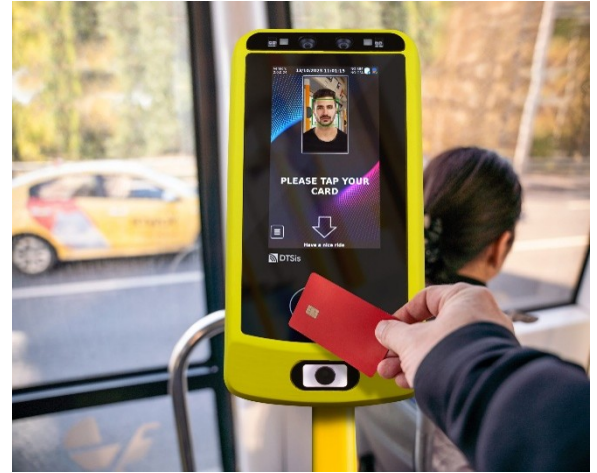


Fig. 4. RFID READER

C. ARDUINO UNO

The Arduino Uno microcontroller serves as the central processing unit of the system. It is based on the ATmega328P chip and provides 14 digital input/output pins, six analog inputs, and a simple USB interface for programming. The Arduino receives the UID from the RFID reader and executes the programmed logic to determine whether the card is valid, whether the balance is sufficient, and what action to perform.

next. Its ease of programming through the Arduino IDE makes it highly suitable for academic and prototype-level projects.

In this system, the Arduino not only processes card data but also controls the output devices. It sends signals to the LCD to display status messages, triggers the ISD1820 voice module to play confirmation sounds, activates the buzzer during errors, and controls the servo motor for gate movement. By acting as the brain of the entire system, the Arduino ensures smooth integration between all hardware components.

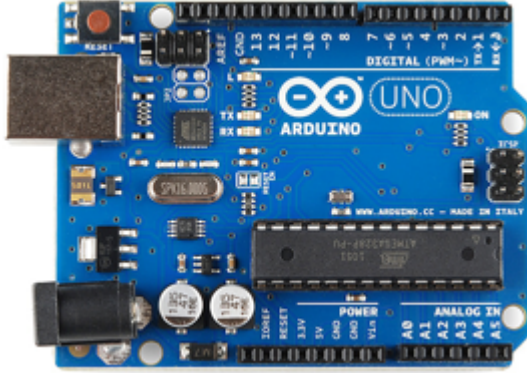


Fig. 5. ARDUINO UNO

D. LCD DISPLAY

The 16x2 Liquid Crystal Display (LCD) provides real-time visual feedback to passengers and operators. It can display up to 32 characters across two lines, making it suitable for short status messages such as “Card Detected,” “Fare Deducted,” or “Invalid Card.” In this project, the LCD is interfaced with the Arduino using digital pins, and its messages are updated dynamically based on the transaction outcome. This instant feedback improves transparency and allows passengers to confirm their transaction on the spot.



Fig. 6. LCD DISPLAY

Beyond passenger communication, the LCD is also useful for debugging and monitoring during system testing. Developers and operators can observe transaction data, balance values, and system errors directly from the screen. This dual role

enhances both usability and maintainability of the system. Due to its low power consumption, affordability, and ease of interfacing, the 16x2 LCD is an ideal choice for real-time display in embedded applications such as smart ticketing.

E. ISD1820 VOICE MODULE

The ISD1820 voice module is responsible for audio confirmation in the ticketing system. It is capable of recording and playing back up to 10 seconds of sound, stored in its on-board non-volatile memory. This makes it suitable for simple voice prompts such as “Ticket Accepted” or “Invalid Card.” The module is interfaced with the Arduino, which triggers playback whenever a specific event occurs. For example, after a successful transaction, the Arduino activates the module to play the confirmation message.

The inclusion of the ISD1820 enhances system accessibility by providing feedback for visually impaired passengers who cannot rely on the LCD display. It also adds a layer of user engagement by making the system more interactive. Since the module is simple to use and requires minimal coding effort, it is ideal for this project. Its compact design, integrated microphone, and speaker compatibility make it an effective and affordable solution for providing real-time voice alerts.



Fig. 7. ISD1820 VOICE ANNOUNCE SYSTEM

F. SERVO GATE

The servo motor in this project acts as the automatic gate mechanism. A servo motor is a type of actuator that provides precise angular motion, typically controlled by Pulse Width Modulation (PWM) signals. In the bus ticketing system, the servo is connected to the Arduino, which controls its rotation based on the transaction outcome. When a valid card is detected and fare is deducted, the servo rotates to open the gate for a few seconds before returning to its closed position.

Using a servo motor ensures controlled and secure passenger entry, preventing unauthorized access. Unlike a DC motor, the servo provides high accuracy and repeatability, making it suitable for applications where exact positioning is required. Its compact size, low cost, and low power consumption make it ideal for small-scale embedded systems. In this project, the servo plays a vital role in physically implementing access control, making the system practical and user-friendly.



Fig. 8. SERVO GATE

G. BUZZER

The buzzer provides simple but essential auditory alerts within the system. It is a small electronic component that produces sound when current is applied. In this project, the buzzer is activated during error scenarios such as invalid card detection or insufficient balance. Its sharp, high-pitched sound immediately notifies passengers and operators of a failed transaction. This ensures that issues are recognized instantly without relying solely on the LCD display.



Fig. 9. BUZZER

The buzzer enhances the reliability of the system by providing multi-sensory feedback. It works in coordination with the LCD and ISD1820 voice module to create a comprehensive alert mechanism. Its low cost, durability, and very simple interfacing with Arduino make it a standard choice in embedded systems. In the context of this project, the buzzer acts as a quick and efficient error notification tool, improving both usability and security of the fare collection system. The buzzer is used for security alert if the transaction is successful or not for confirmation user and conductor.

VII. INTEGRATION PHASE

In the integration phase, it combines the hardware components and software logic into a single, fully functional prototype. This stage ensures that all modules work together seamlessly and that the system performs as expected under real operating conditions. The process begins with hardware integration, where all individual components – RFID reader, LCD display, servo motor, buzzer, and ISD1820 voice module – are connected to the Arduino Uno according to the circuit diagram. Special attention is paid to proper power distribution to ensure a stable 5V supply for sensitive components such as the RFID reader and the servomotor. The wiring is neatly arranged to avoid interference and short circuits, and the breadboard or PCB connections are verified for continuity.

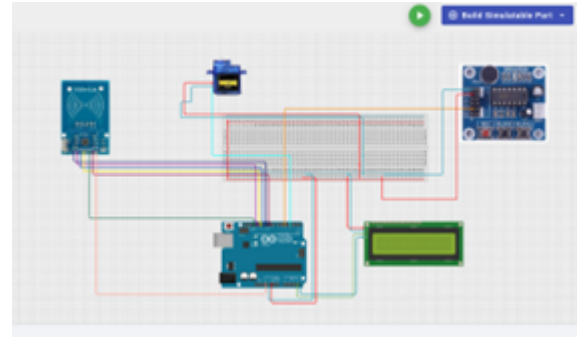


Fig. 10. INTEGRATION PHASE

Once the hardware setup is complete, the software program developed in the Arduino IDE is uploaded to the microcontroller. Integration testing begins with powering up the system and checking the initialization of each component. The LCD is verified for correct startup messages, the RFID reader is tested for card detection, and the servo motor is checked for smooth gate operation.

After verifying individual components, system-level integration is performed. Test cards are tapped to ensure correct communication between the RFID reader and Arduino, proper fare deduction, and simultaneous activation of LCD, servo, buzzer, and voice module. Multiple scenarios such as valid card, low balance, and invalid card are simulated to confirm that the logic flow handles each case appropriately. Debugging is done using the Arduino Serial Monitor to observe internal data such as card IDs and balance updates in real time. Any timing mismatches, delays, or wiring issues are corrected during this phase.

Debugging is done using the Arduino Serial Monitor to observe internal data such as card IDs and balance updates in real time. Any timing mismatches, delays, or wiring issues are corrected during this phase. By the end of the integration phase, the system functions as a unified unit, with synchronized input processing, decision-making, and output response. This phase confirms the readiness of the project for final testing, performance measurement, and real-world demonstration, ensuring

it meets the objectives of fast, accurate, and automated fare collection for public transport.

VIII. TESTING PHASE

The testing phase is the final and most crucial step in the development of the RFID-based fixed fare bus ticketing system. This phase ensures that the entire system operates correctly, reliably, and consistently under different operating conditions. The objective is to verify hardware functionality, validate software logic, and measure system performance to ensure it meets project requirements.

Testing begins with unit testing, where each hardware component is tested individually. The RFID reader is verified for accurate detection of card IDs, the LCD display is tested for proper message display, the servo motor is checked for smooth gate movement, and the buzzer and ISD1820 voice module are tested for correct alerts and voice playback. Unit testing ensures that each component is working as expected before full system integration.

Next, integration testing is performed to check the interaction between components. Sample RFID cards are tapped to ensure that the Arduino receives data from the reader, processes the information, deducts the fixed fare, and activates the correct outputs. During this stage, timing issues such as gate delay, LCD refresh speed, and buzzer duration are fine-tuned for smooth operation.

Once integration is verified, system testing is conducted under real operating conditions. Multiple scenarios are tested, including valid card transactions, invalid card detection, and insufficient balance cases. For valid transactions, the LCD must display "Fare Deducted," the voice module must play a confirmation message, and the servo gate must open within the expected time. For invalid scenarios, the buzzer must activate and the gate must remain closed.

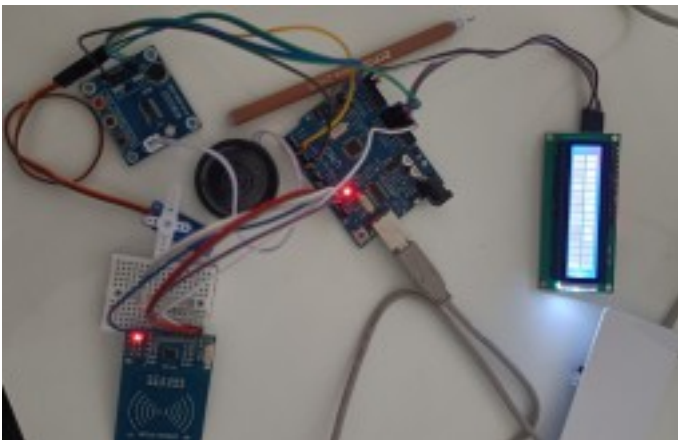


Fig. 11. TESTING PHASE

Finally, performance testing is carried out to measure important metrics such as detection accuracy, system response time, and reliability under continuous use. Multiple cards are tapped in quick succession to test the system's ability to handle

peak passenger loads. Any errors or failures are recorded and corrected.

REFERENCES

- [1] Oudah, A. (2016). RFID-based automatic bus ticketing: features and trends. IOP Conference Series: Materials Science and Engineering, 114(1), 012146. IOP Publishing.
- [2] Priyanka, S., Abhishek, R., Pasha, A., Bhat, M. C., and Raj, D. D. (2020). Smart ticket system for metro train.
- [3] Subramaniyan, J., Kumar, P. S., Sripriya, S., Jegan, C., and Jenny, M. (2021). Smart Ticketing System for Public Transport Vehicles. Recent Trends in Intensive Computing.
- [4] Hasan, Tangim, Golam and Islam, Kafiul and Khandokar Md, Alam, Arif, 2010. "RFID-based ticketing for public transport systems ", International Journal of Engineering Research and Technology.
- [5] Dr. Chowdhury, Prasun Bala, Poulami Addy, Diptadeep and Giri, Sumit and Chaudhuri, Aritra. 2016, "RFID and Android based smart ticketing and destination announcement system ". 2020 5th International Conference on Communication and Electronics System.
- [6] Rahul S, Manushree B, Bharath T and Anvesh P, June - 2022, "RFID Based Smart Master Card For Bus Train Metro Ticketing ", International Research Journal of Modernization in Engineering Technology and Science.
- [7] Piyush C, Rakesh K. K, Prakhar S, May 2014, "RFID-based ticketing for public transport systems ", February 2016- IOP Conference Series Materials Science and Engineering.
- [8] Saurabh C, Balram T, 2014, "Public transport system ticketing system using RFID and ARM processor ", Perspective Mumbai bus facility, International Journal of Electronics and Computer Science Engineering.